Towards Human-centered AI

Understanding and Shaping Human Decisions with AI

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Decision Making

Healthcare (e.g., Nature Communications 2025, FAccT 2025, ML4H 2023)

Persuasion (e.g., ACL 2023, TACL 2019, WWW 2016)

Political communication (e.g., PNAS Nexus 2024, Findings of EMNLP 2023, ICWSM 2016)

Scientific Discovery

AI scientists (e.g., NeuriCo, VERITAS, ACL 2025)

Interdisciplinary collaboration (sciences, social sciences) (e.g., LivingScience, NeuriCo-battery, Virtual Cell Challenge, ICML Workshop 2026)



Understanding and Improving AI

Causal reasoning (e.g., NeurIPS 2026a, ICLR 2026, TMLR 2024)

Interpretability/explanations (e.g., NeurIPS 2026b, TMLR 2023, FAccT 2020)

AI alignment/safety (e.g., ICML 2026, EMNLP 2026, NeurIPS 2025)

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Policy outcomes hinge on human decisions



Understanding human decisions

Chao-Chun Hsu, Ziad Obermeyer, Chenhao Tan
Nature Communications, 2025



Shaping human decisions

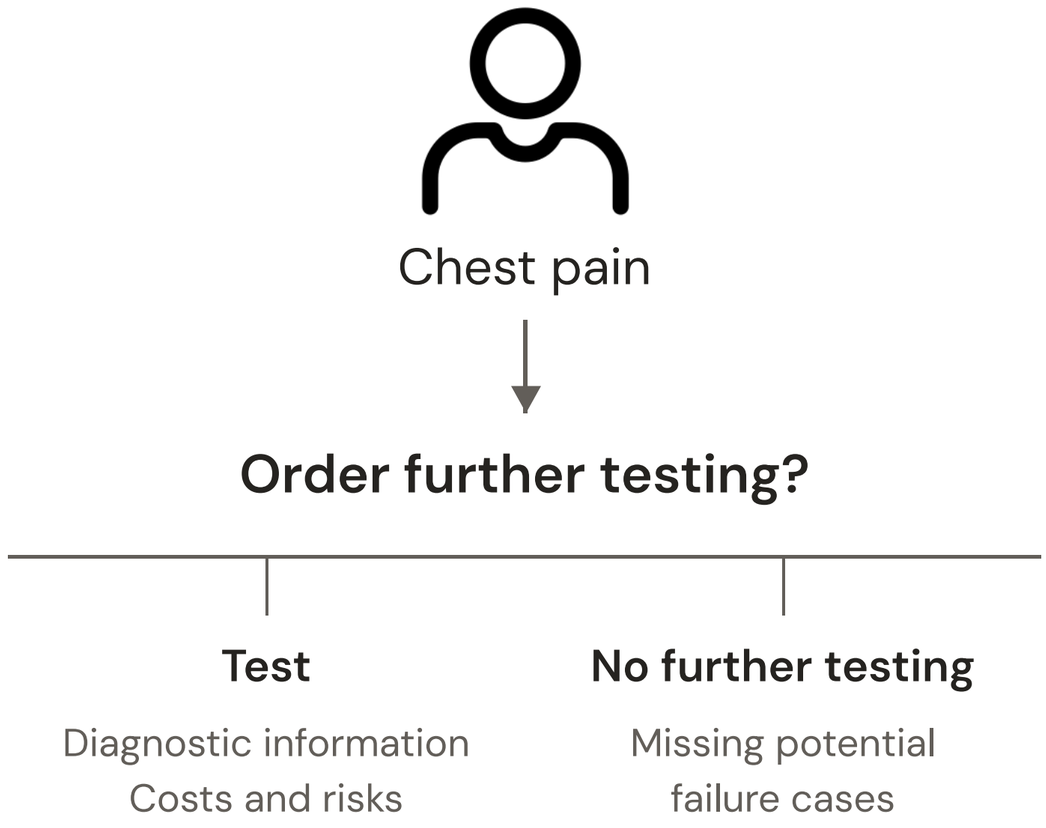
Chacha Chen, Shi Feng, Amit Sharma, Chenhao Tan · TMLR / FAccT 2023
Han Liu et al. · ICLR 2023

Decision of Interest in Emergency Care

A patient arrives with chest pain.

The physician must decide whether to order further testing for acute coronary syndromes (“heart attack”).

Testing low-risk patients can impose costs and risks with little clinical benefit.



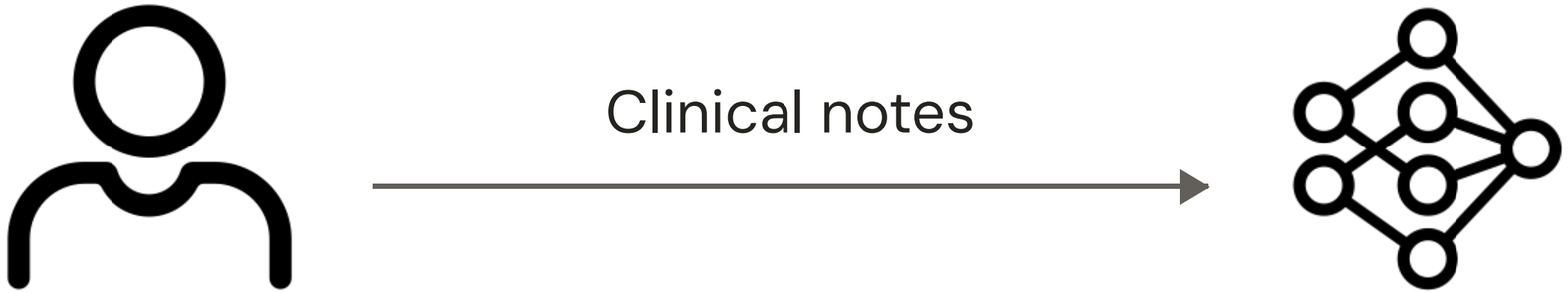
Does *physician fatigue* relate to decision quality?

Our key contribution is to propose a new measurement of fatigue and connect the measured fatigue with decision quality.

The Paradox of Physician Fatigue

- **Physician burnout is prevalent.**
  - In a 2022 report, 65% of emergency physicians reported burnout, the highest rate of any medical specialty. (Carbajal 2022)
- **Earlier research found limited evidence connecting fatigue with real patient outcomes.** (Arnedt et al. 2005; Ayas et al. 2006; Barger et al. 2006; Johns et al. 2009; Landrigan et al. 2004; Lockley et al. 2004)
- One possible reason: studies used coarse measures related to workload.

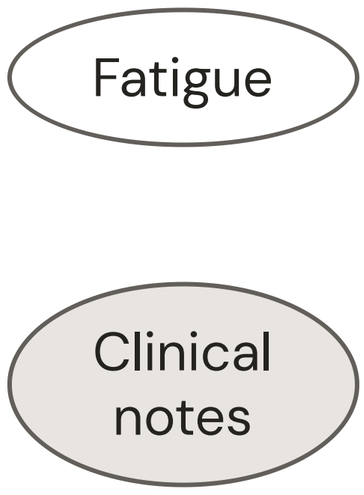
Can we learn about physician fatigue from clinical notes?



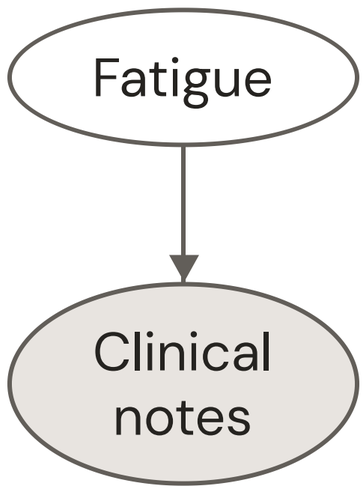
Example clinical notes

> Chenhao is a 34-year-old male who presents with a chief complaint of chest pain for the past 3 days. The pain is described as a dull, pressure-like sensation located in the mid-sternal area, radiating to the left shoulder and arm. It is rated as 6/10 in intensity and worsens with physical activity, such as climbing stairs, but improves with rest. He denies any recent trauma to the chest.

Intuitively, workload should not affect clinical notes in the ideal world



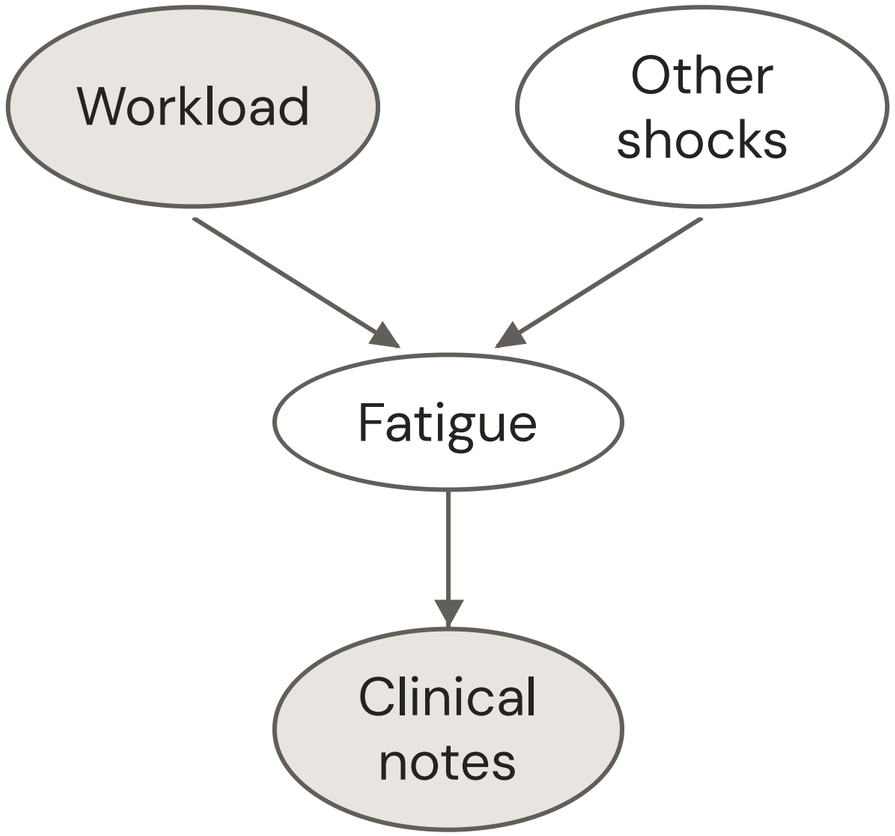
However, physicians do get tired



Fatigue may affect the notes physicians write.

The challenge: we do not have ground-truth measurements of physician fatigue.

From Workload to a Note-Based Fatigue Measure



Assumption: workload affects notes through physician fatigue.

Train a model to predict prior workload from a note.

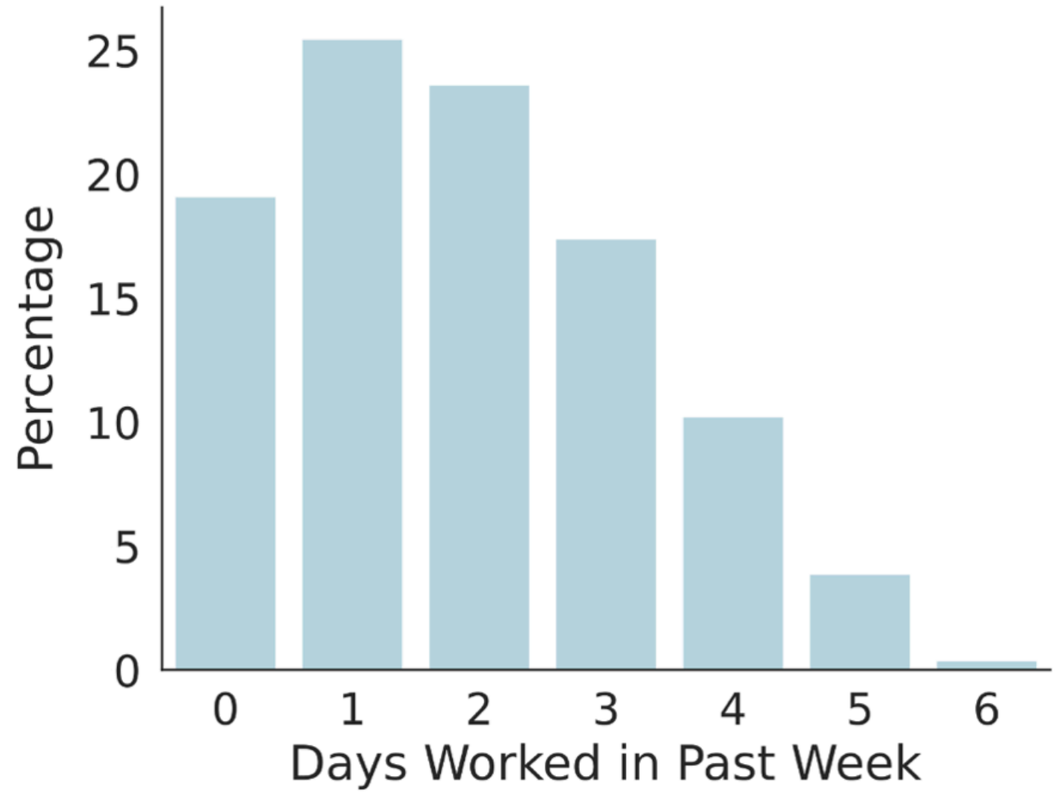
Its predictions will capture fatigue, including those caused by other shocks (see proof in the paper).

Data and Prediction Task

129,228 emergency department encounters

60 physicians working **11,592 shifts**

Predict high versus low prior workload using the text of a clinical note.

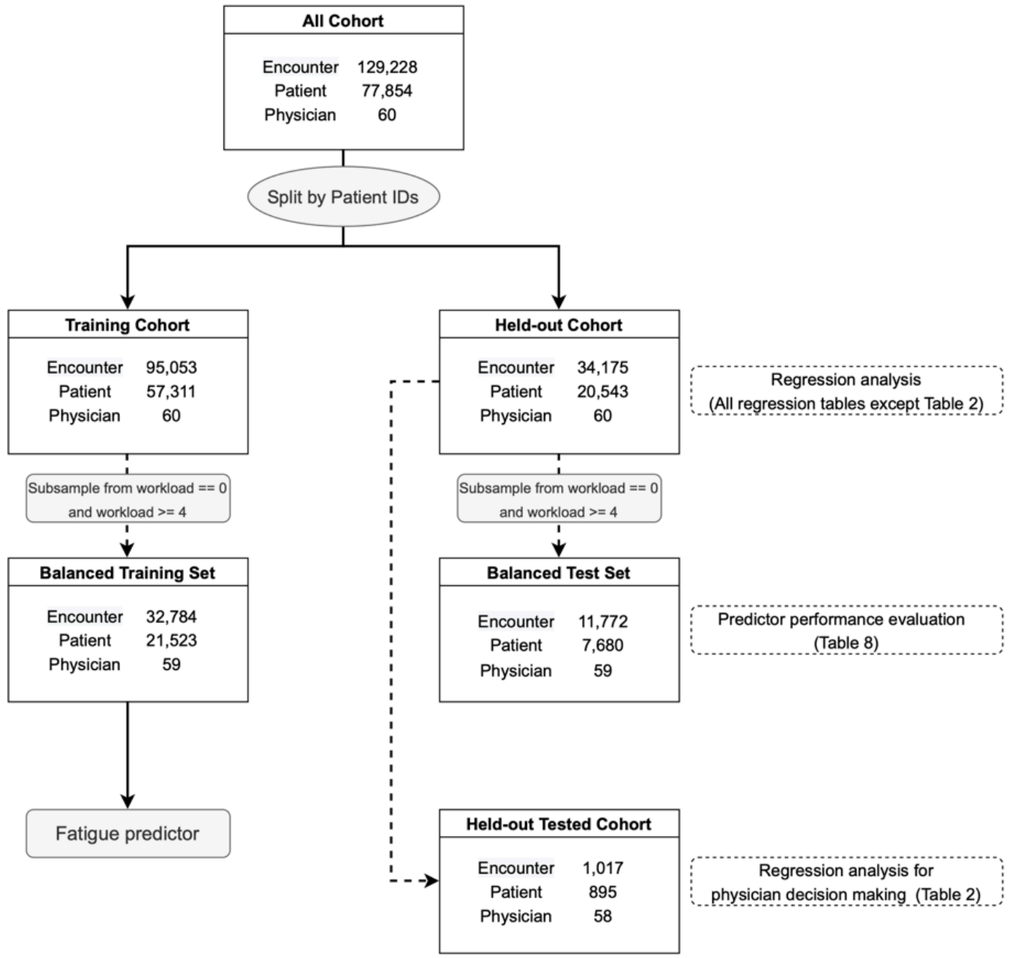


Prior workload in the supplied study slides

Data and Prediction Task

- **X:** a clinical note
- **Y:** high versus low prior workload

Sanity check: patient characteristics do not differ with prior workload.



Features

- **Note length**
- **Note predictability:** average negative log probability from fine-tuned GPT-2
- **Note readability**
- **Lexicon counts:** pronouns, affect, and cognitive processes (LIWC)
- **Clinical language:** medical words from SNOMED and stopwords

$$p(w_1, \ldots, w_N) = \prod_{t=1}^{N} p(w_t \mid w_{<t})$$

$$-\frac{1}{N} \sum_{t=1}^{N} \log p(w_t \mid w_{<t})$$

Average negative log probability (log perplexity)

Notes Predict Prior Workload

Input / method	AUC-ROC (%)
Chief complaint baseline	50.0
Zero-shot Vicuna-7B	53.2
Clinical note features	**60.1**

Predicting workload is the starting point for validating the fatigue measure.

Other Measures of Fatigue

Hypothesis: if notes capture fatigue, predicted fatigue should also track other measures of fatigue.

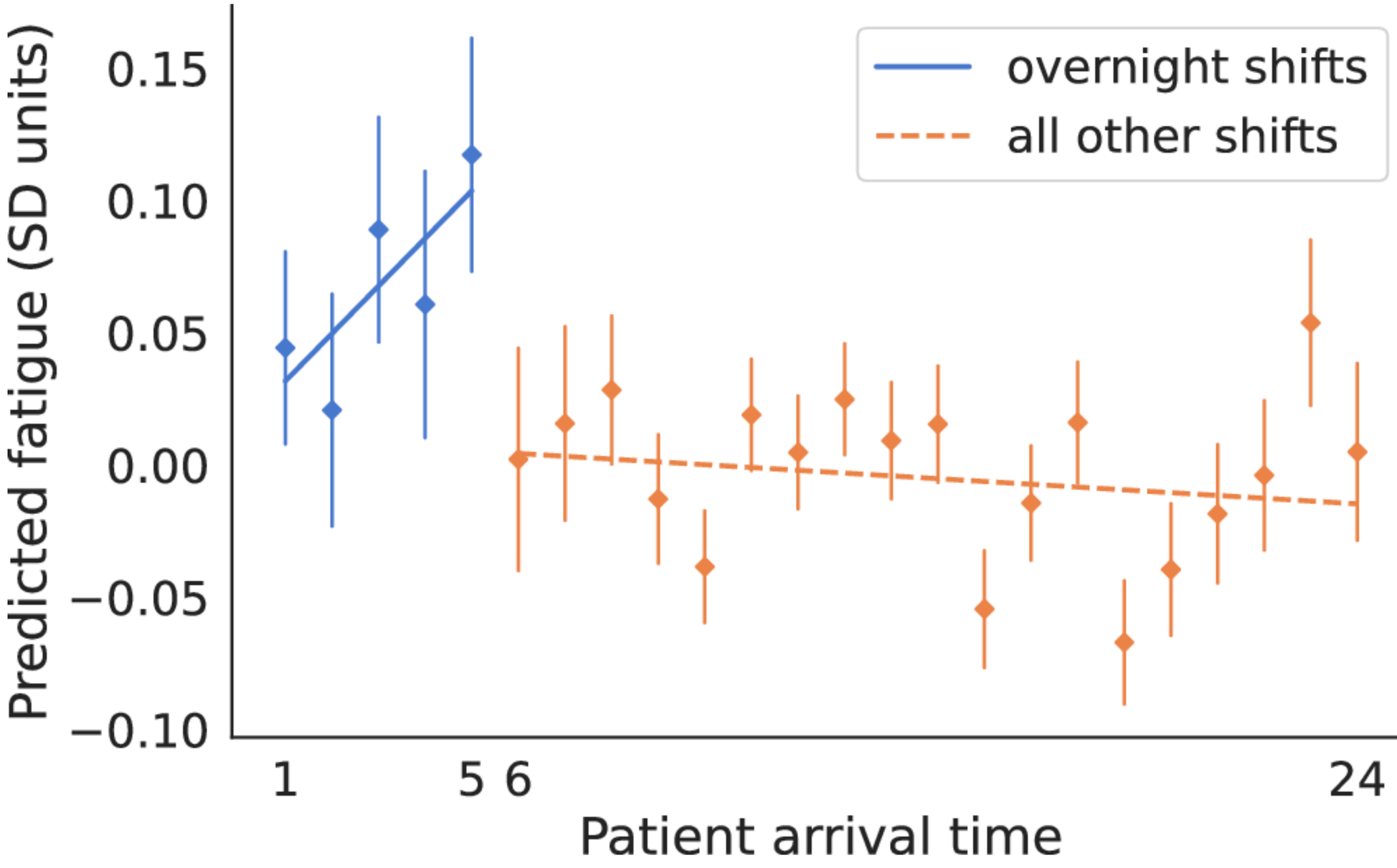
Measure	Definition
Overnight shift	Patient arrives between 1:00 a.m. and 5:59 a.m.
Circadian disruption	Variance of physician shift start times over the week
Patients seen prior	Order of the physician encounter within a shift

Predicted Fatigue Correlates with Other Fatigue Measures

	(1)	(2)	(3)	(4)
	Hospital Dataset			
	Workload (Day)	Overnight Shift	Variation in Shift - Start-Time	Patients Seen Prior
Fatigue	0.384***	0.041*	5.606*	1.073***
	(0.087)	(0.018)	(2.265)	(0.298)
Intercept	3.072***	0.158***	29.855***	9.011***
	(0.102)	(0.019)	(2.669)	(0.352)
Controls				
Time of day	YES	NO	YES	YES
Day of week	YES	YES	YES	YES
Week of year	YES	YES	YES	YES
Year	YES	YES	YES	YES
Demographics	YES	YES	YES	YES
Chief complaint	YES	YES	YES	YES
Physician	YES	YES	YES	YES
Length of stay	YES	YES	YES	YES
Insurance type	YES	YES	YES	YES

$p < 0.05^{*}, p < 0.01^{**}, p < 0.001^{***}$

Predicted Fatigue Rises Overnight



External Validation on MIMIC-III

	(1)	(2)	(3)	(4)	(5)
	Hospital Dataset				*Ext. Validation*
	Workload (Day)	Overnight Shift	Variation in Shift - Start-Time	Patients Seen Prior	Overnight Shift
Fatigue	0.384^{***}	0.041^{*}	5.606^{*}	1.073^{***}	0.364^{*}
	(0.087)	(0.018)	(2.265)	(0.298)	(0.162)
Intercept	3.072^{***}	0.158^{***}	29.855^{***}	9.011^{***}	-0.239
	(0.102)	(0.019)	(2.669)	(0.352)	(0.288)
Controls					
Time of day	YES	NO	YES	YES	YES
Day of week	YES	YES	YES	YES	N/A
Week of year	YES	YES	YES	YES	N/A
Year	YES	YES	YES	YES	N/A
Demographics	YES	YES	YES	YES	YES
Chief complaint	YES	YES	YES	YES	YES
Physician	YES	YES	YES	YES	YES
Length of stay	YES	YES	YES	YES	YES
Insurance type	YES	YES	YES	YES	YES

$p < 0.05^{*}, p < 0.01^{**}, p < 0.001^{***}$

Fatigue and Testing Decisions

Decision: whether to test for acute coronary syndromes

Outcome: the fraction of tests that are positive

Compare this measure of decision quality with note-based fatigue and with prior workload.

Predicted Fatigue and Testing Yield

	Effect of workload on yield of testing	Effect of predicted fatigue on yield of testing
Coefficient	−0.0081	−0.2935*
	(0.009)	(0.147)
Controls		
Time of day	YES	YES
Day of week	YES	YES
Week of year	YES	YES
Year	YES	YES
Demographics	YES	YES
Chief complaint	YES	YES
Physician	YES	YES
Length of stay	YES	YES
Insurance type	YES	YES

$n = 1{,}017$; $p < 0.05^*$

Prior workload alone: no detectable relationship with testing yield.

19% lower testing yield for each one standard deviation increase in predicted fatigue

Fatigue Scores and Patient Race

	Workload (Day)	Predicted Fatigue
Race (vs. White)		
Black	−0.0009	0.0112***
	(0.017)	(0.001)
Hispanic	−0.0125	0.0192***
	(0.018)	(0.001)
Other	−0.0213	0.0086***
	(0.027)	(0.002)
Female (vs. Male)	0.0041	−0.003**
	(0.015)	(0.001)
Age	0.0006	−0.0003***
	(0)	(0.00003)
Intercept	3.2354***	0.569***
	(0.084)	(0.005)
Controls		
Time of day	YES	YES
Day of week	YES	YES
Week of year	YES	YES
Year	YES	YES
Chief complaint	YES	YES
Physician	YES	YES

$p < 0.05^{*}, p < 0.01^{**}, p < 0.001^{***}$

Prior workload does not differ by patient race.

Notes about Black and Hispanic patients show higher predicted fatigue than notes about White patients.

What Changes in the Notes?

Feature	Correlation with physician workload
Word Un-predictability	
Perplexity (log)	-0.092^{***}
Cognitive words (LIWC)	
Insight (fraction)	-0.09^{***}
Certainty (fraction)	0.075^{***}
Pronouns (LIWC)	
First person singular pronouns (fraction)	-0.07^{***}
Impersonal pronouns (fraction)	-0.049^{***}
Affect (LIWC)	
Anger (fraction)	0.025^{***}
Readability	
Flesch-Kincaid grade	-0.05^{***}

$p < 0.001^{***}$

Notes written after heavier workload are more predictable.

They use fewer insight words (“think,” “explain”) and more certainty words (“clear,” “never”).

More findings in the paper.

What Happens When AI Writes the Notes?

LLM-generated text is predictable and may receive a high “fatigue” score.

Text source	Score deviation from mean
Greedy generation	+189%
Sampled generation	+8%
Original History of Present Illness	−66%

Automatically generated notes can erase signals from human writing.

Integrating AI into Note Generation

Separate the different goals of note writing.

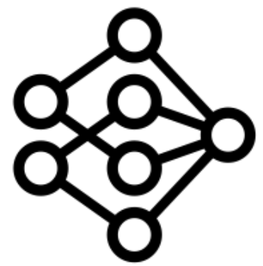
Design AI to help physicians provide better care.

Help physicians identify valuable information to ask for during an encounter.

Understanding human decisions

Chao-Chun Hsu, Ziad Obermeyer, Chenhao Tan
Nature Communications, 2025

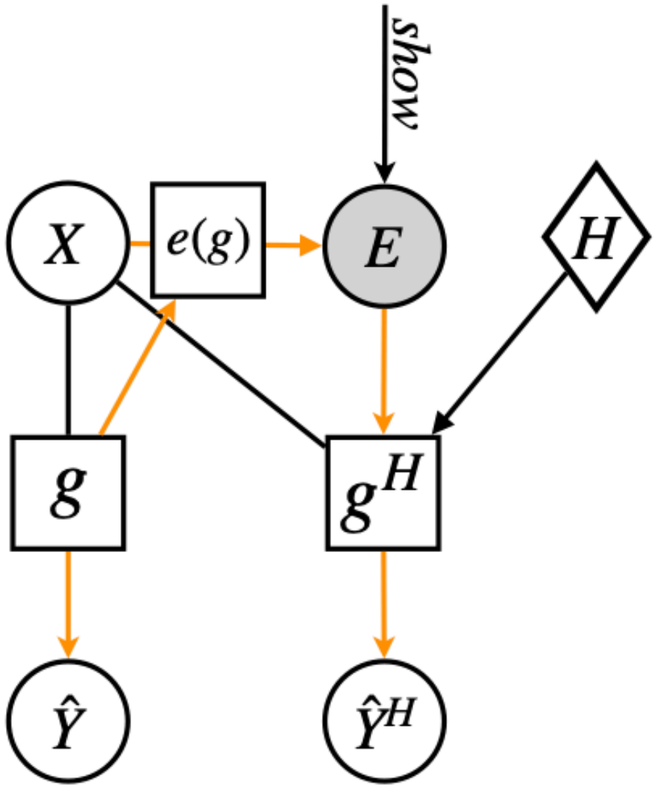




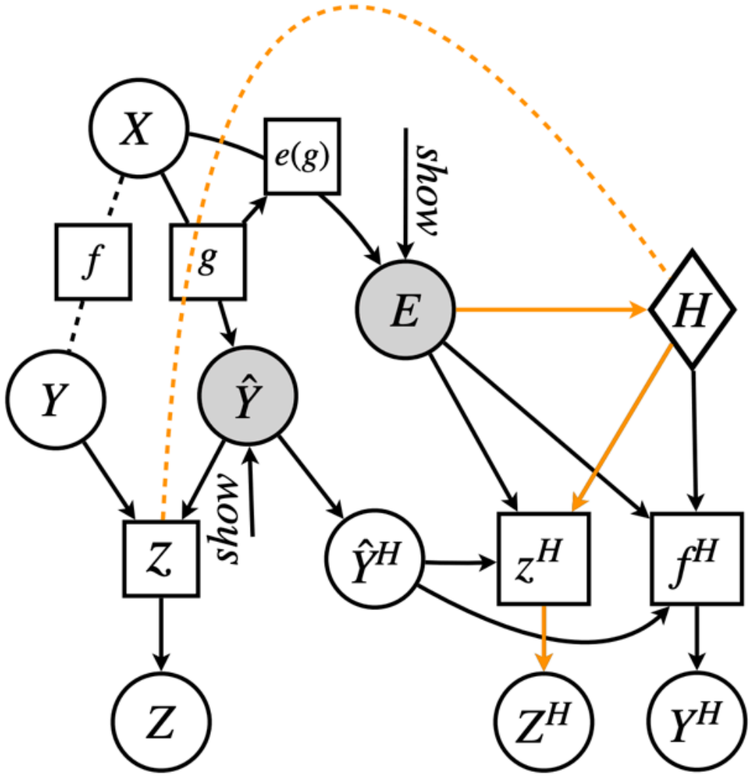
Shaping human decisions

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Han Liu et al. · ICLR 2023

Explanations Need to Activate Task-Specific Intuitions



Understanding model behavior

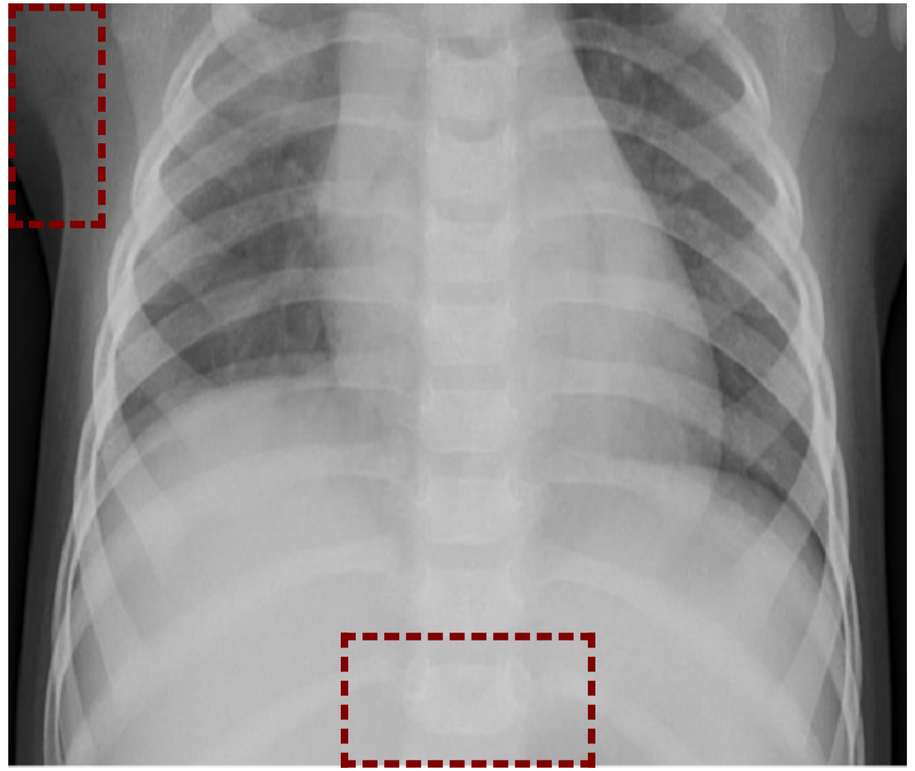


Connecting to the task

Task: Pneumonia Diagnosis

Consider two users: one with task-specific intuitions, and one without.

The same explanation can mean different things to different people.

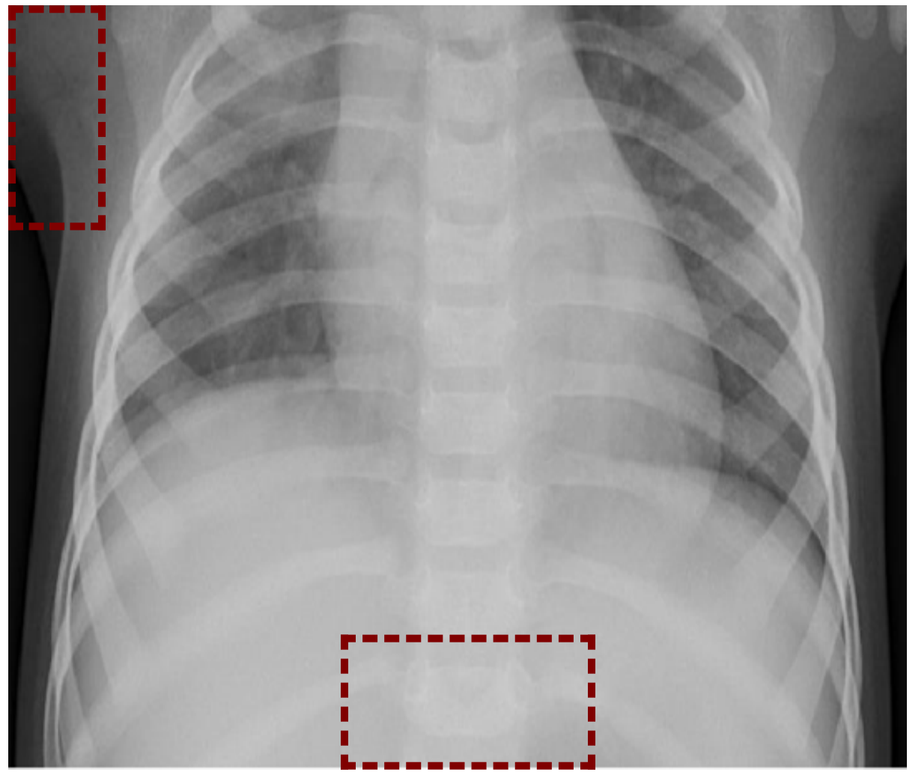


The explanation: regions the AI relies on

Without Task-Specific Intuitions



The user cannot make sense of the highlighted information. Understanding the task remains bounded by understanding the model.

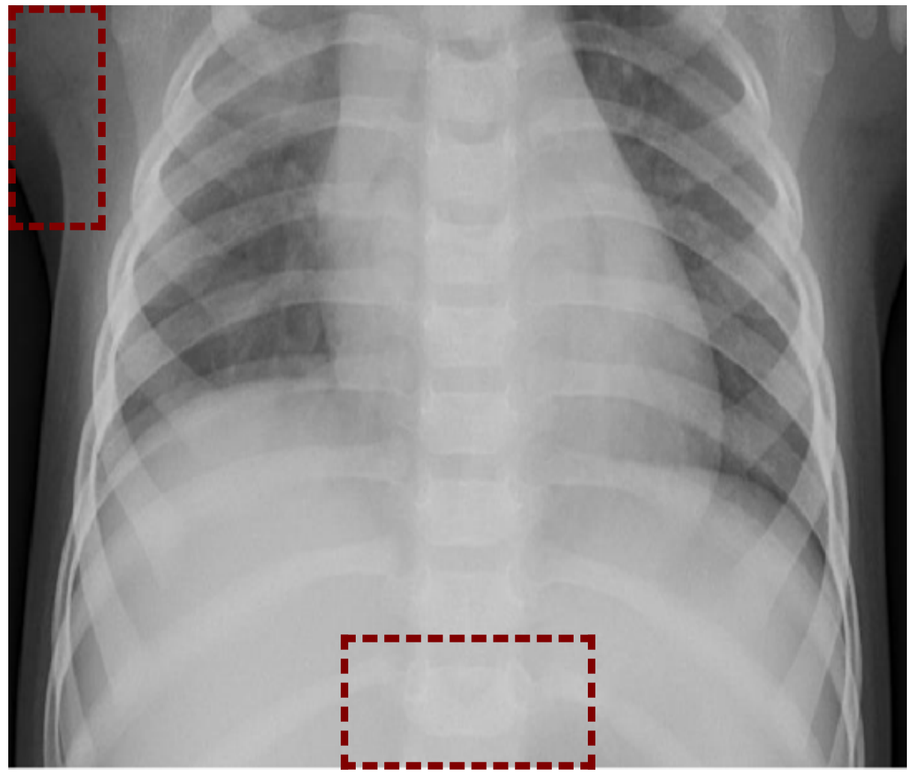


With Task-Specific Intuitions



A user with relevant task knowledge may judge whether the model uses valid information.

This information may help the person judge when the model is right.

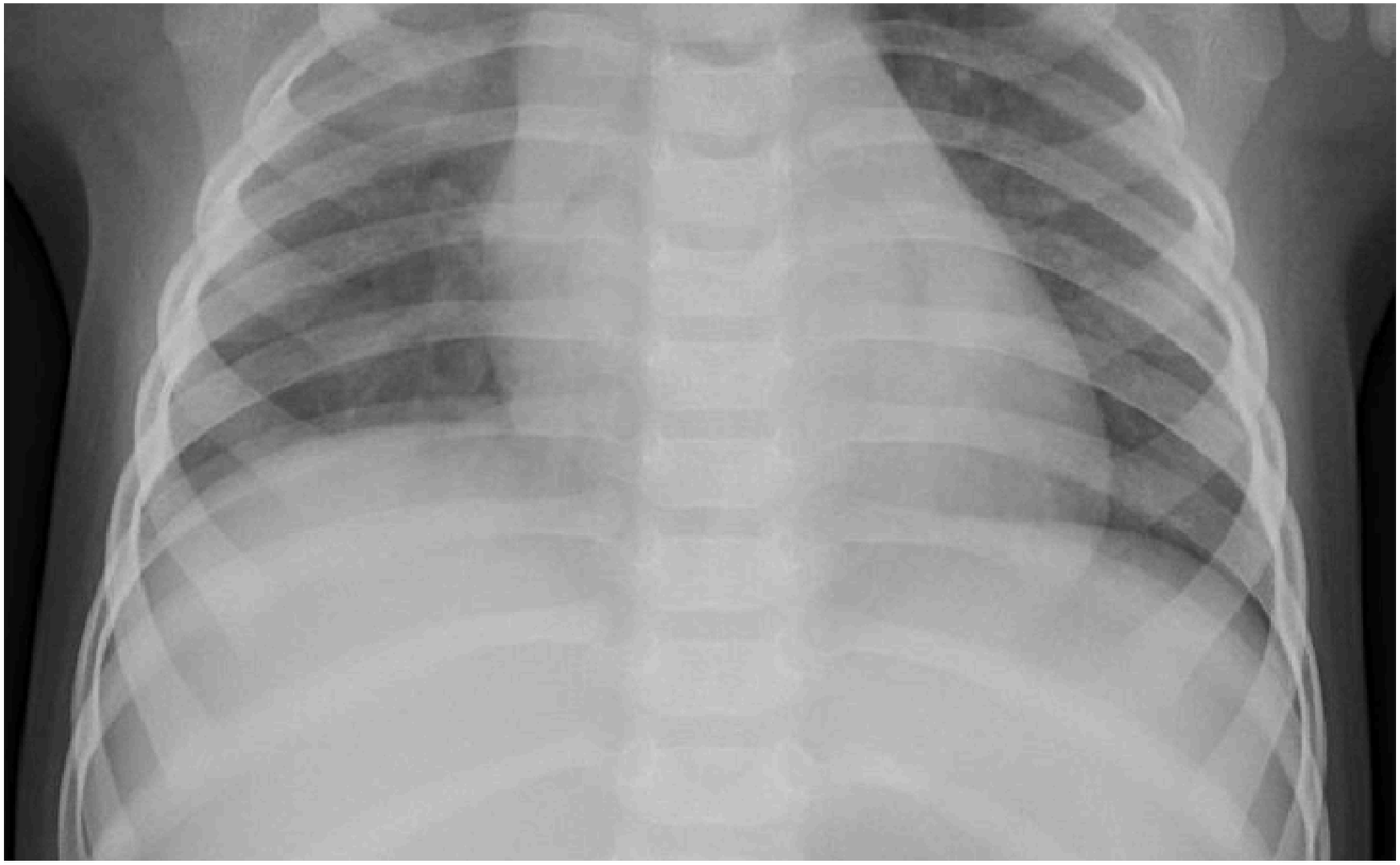


Human-Centered Explanations

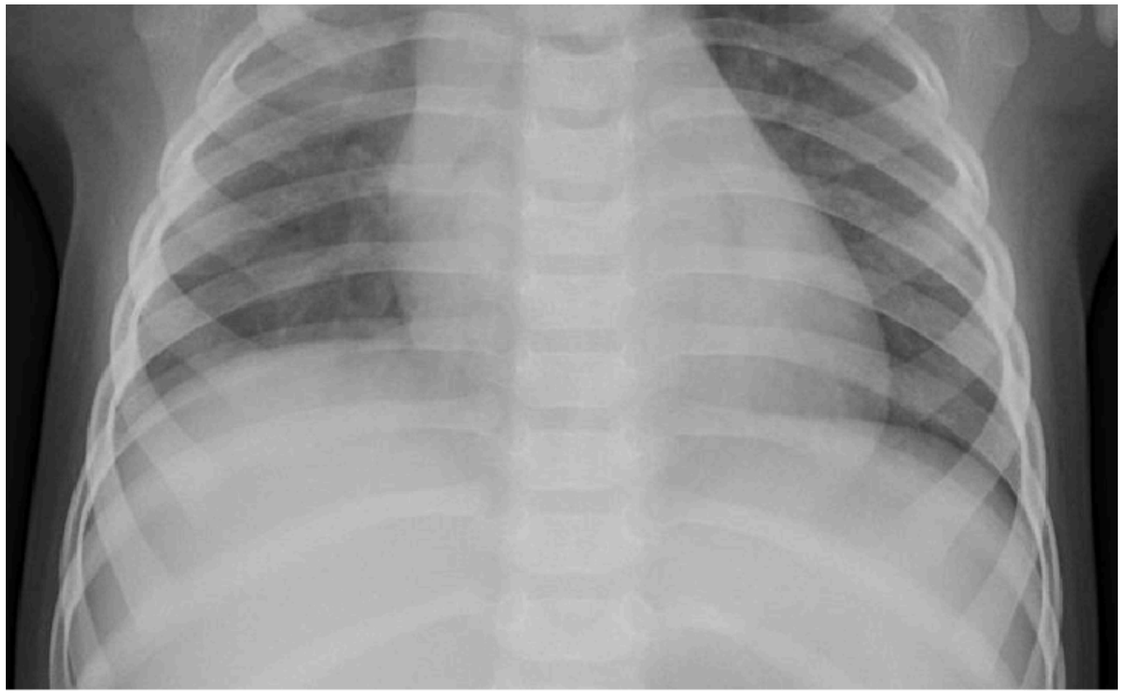
1. Articulate how people may interact with explanations.

2. Generate explanations that fit this mechanism.

Pneumonia Diagnosis



Pneumonia Diagnosis: Supporting the Prediction



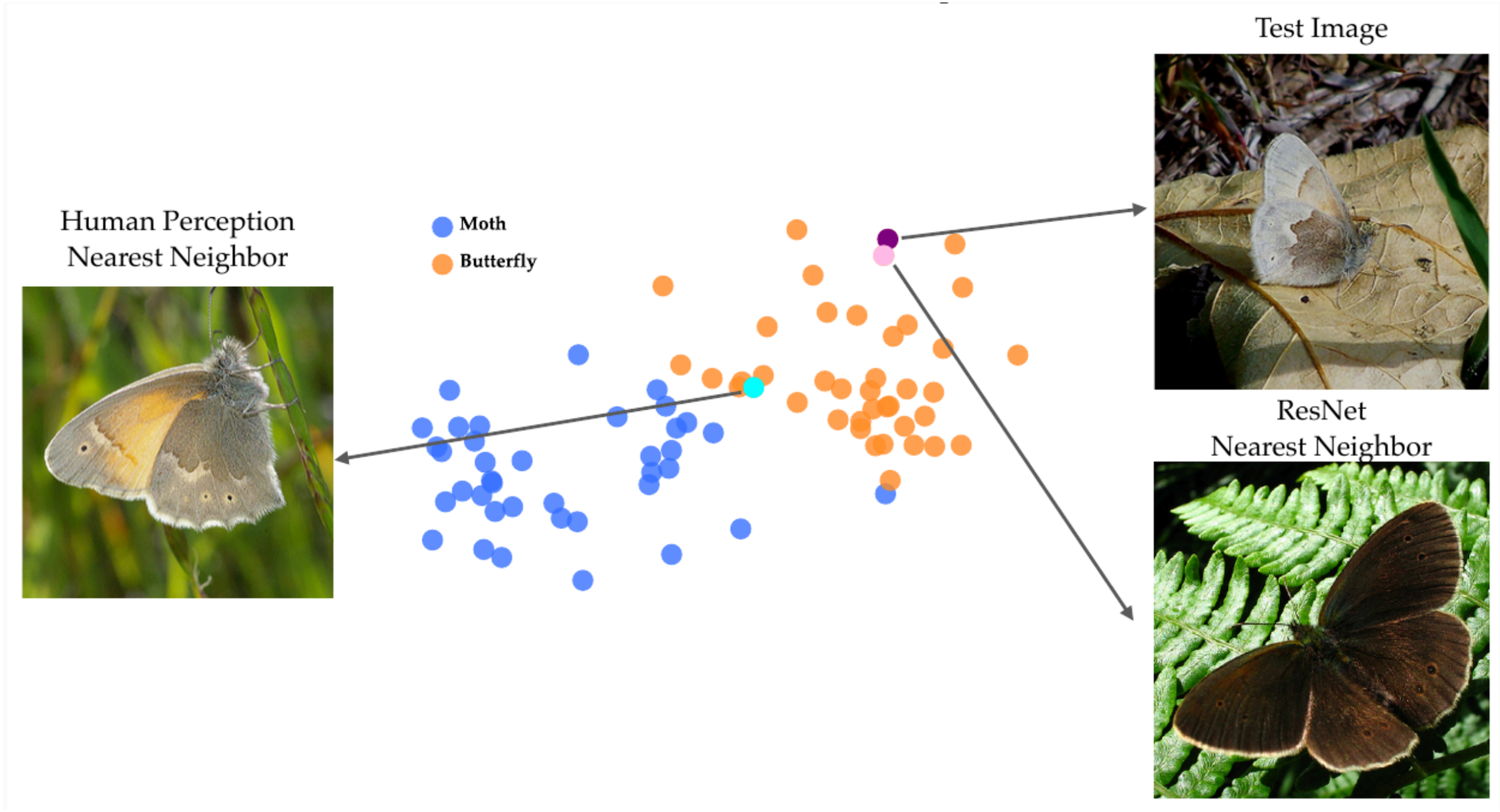
AI predicts pneumonia



Example supporting the prediction

Key assumption: support helps when model similarity connects with the similarity people perceive.

AI Similarity Can Differ from Human Similarity



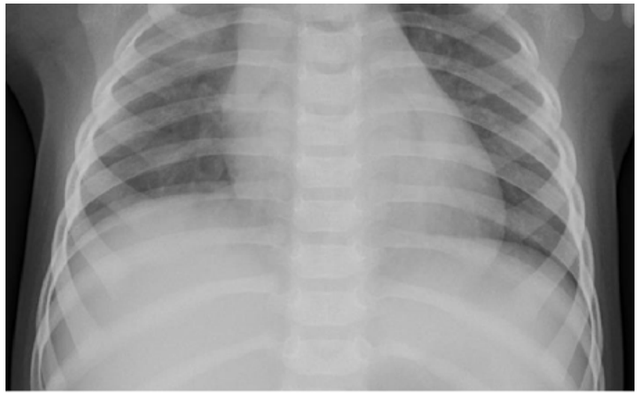
Learning Representations That Match Human Similarity

Case-based explanations are derived from AI representations.

If the representation uses a different notion of similarity, the examples may be unhelpful.

Learn human-compatible representations.

Collect Human Triplet Judgments



Query



Reference A



Reference B

Which reference looks more similar to the query?

Learning Human-Compatible Representations

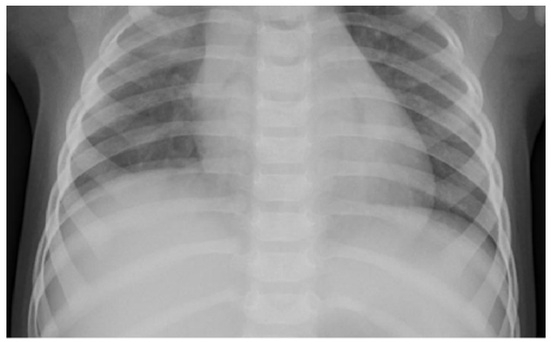
A multi-task objective combines:

Image classification + **Human judgment prediction**

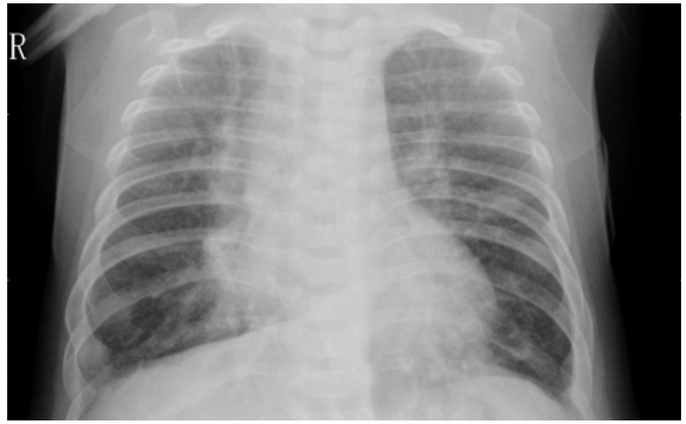
Learn to distinguish the classes — Learn which examples people see as similar

$$\lambda \underbrace{\left[- \sum_{(x,y)\sim D} \log\left(p_\theta(y|x)\right) \right]}_{\text{Cross-entropy loss}} + (1-\lambda) \underbrace{\left[\sum_{(x^r,x^+,x^-)\sim T} \max\left(d_\theta(x^r,x^+) - d_\theta(x^r,x^-) + 1, 0\right) \right]}_{\text{Triplet margin loss}}$$

Experiment Setup: Neutral Decision Support



Test image



Nearest neighbor in the predicted class



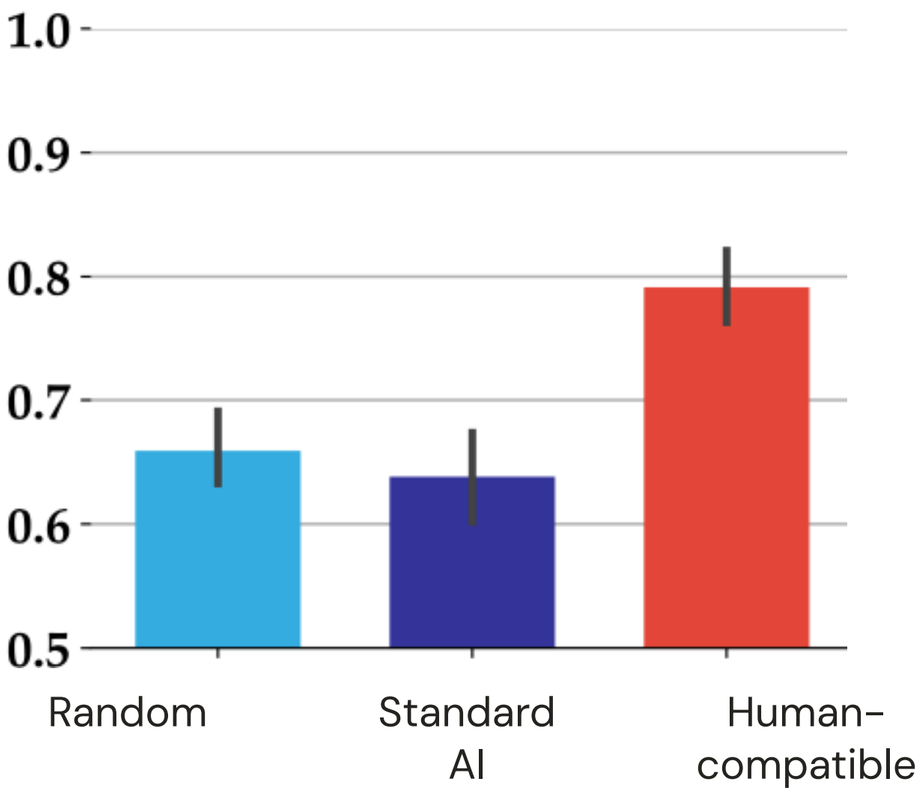
Nearest neighbor in the other class

Choose a diagnosis by comparing the test image with one example from each class.

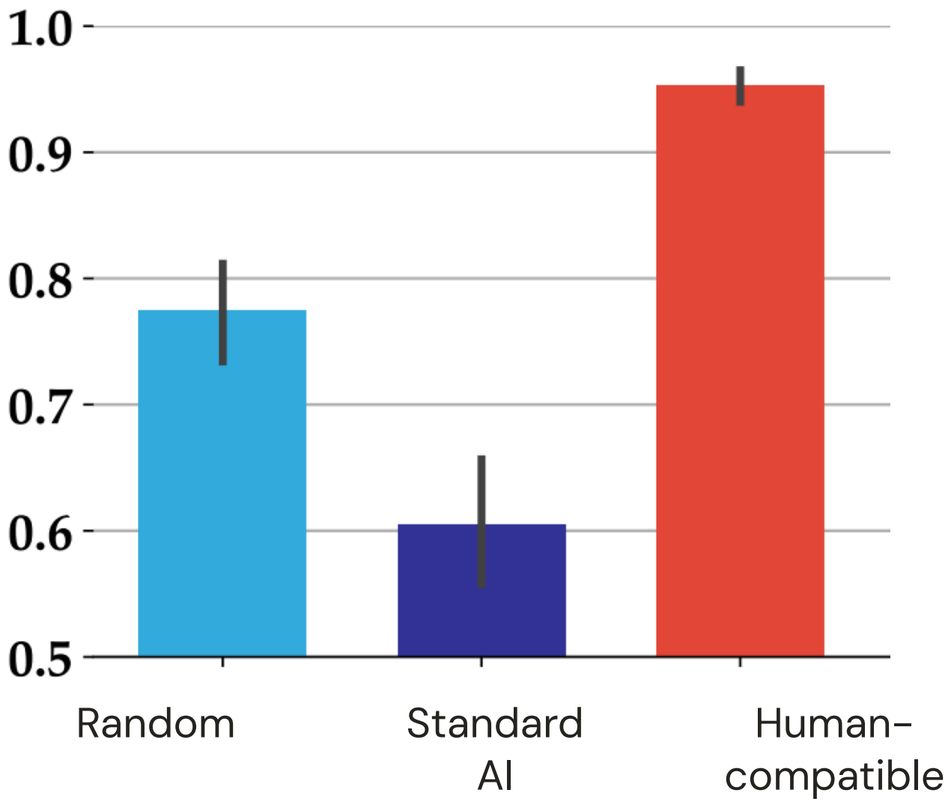
Experiment Setup: Comparing Three Representations

Condition	How examples are chosen
Random	Random supporting examples
Standard AI	Nearest examples in the standard learned representation
Human-compatible AI	Nearest examples in a representation trained with human judgments

Human-Compatible Examples Improve Decision Support



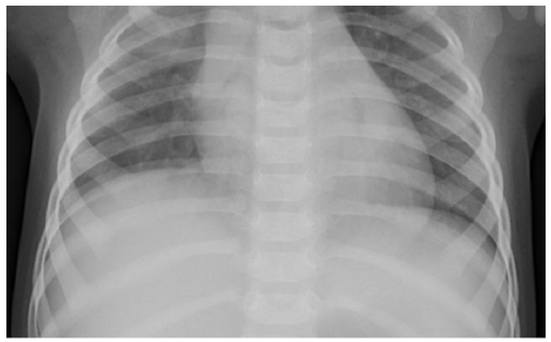
Pneumonia classification



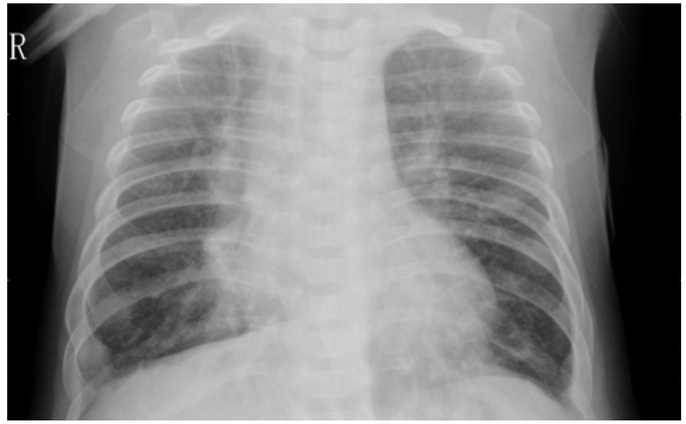
Butterfly vs. moth

Human decision accuracy · error bars show 95% confidence intervals

Experiment Setup: Persuasive Decision Support



Test image



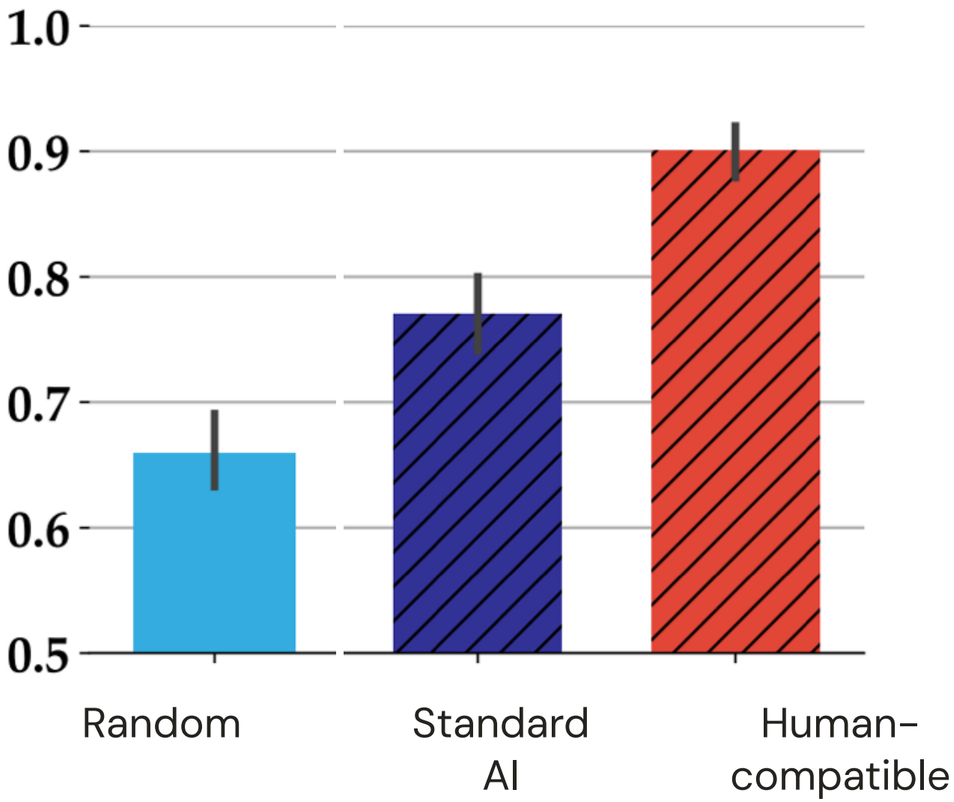
Nearest neighbor in the predicted class



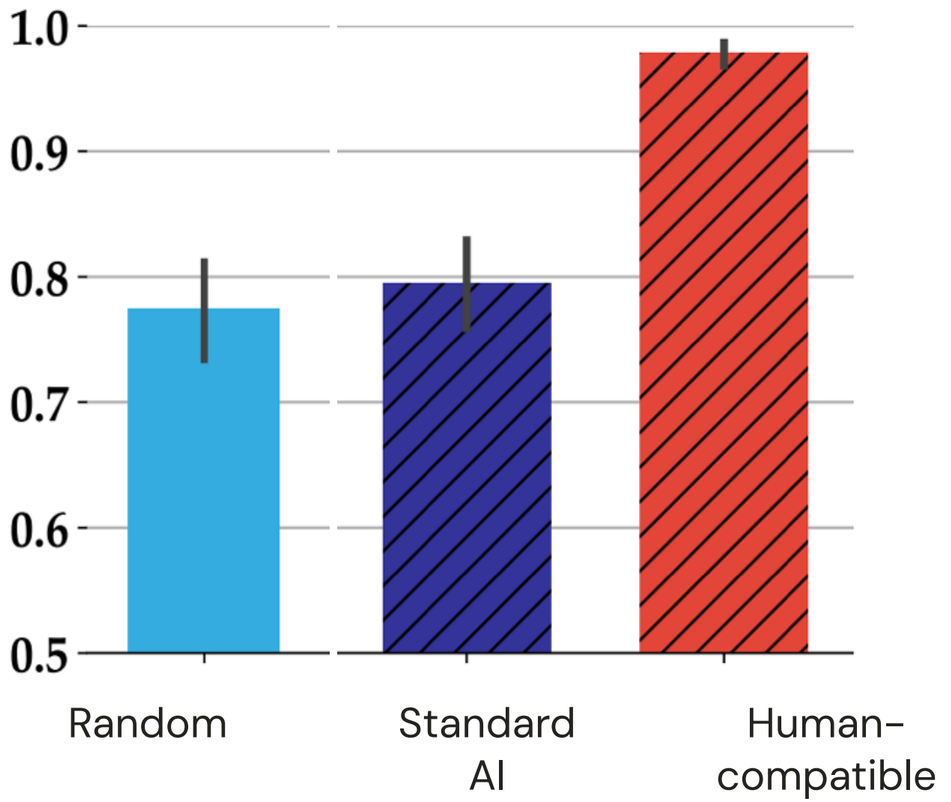
Farthest neighbor in the other class

Make the contrast more favorable to the prediction by choosing a distant example from the other class.

Human-Compatible Examples Are Also More Persuasive



Pneumonia classification



Butterfly vs. moth

Human decision accuracy · error bars show 95% confidence intervals

Future Directions

1. **Interactive intervention**

   AI that helps people reason through policy choices

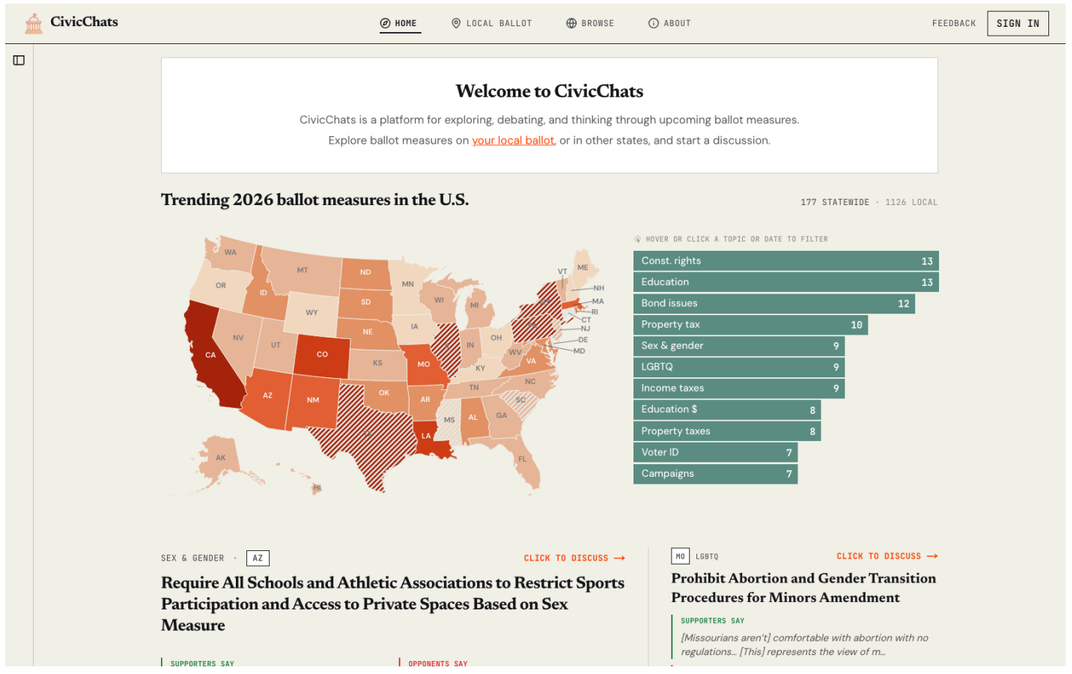
2. **Continuously updated evidence**

   Research findings that stay useful for decision making

3. **AI scientist for AI safety**

   Evaluation that responds to emerging incidents and adapts to model behavior

Interactive Intervention: CivicChats



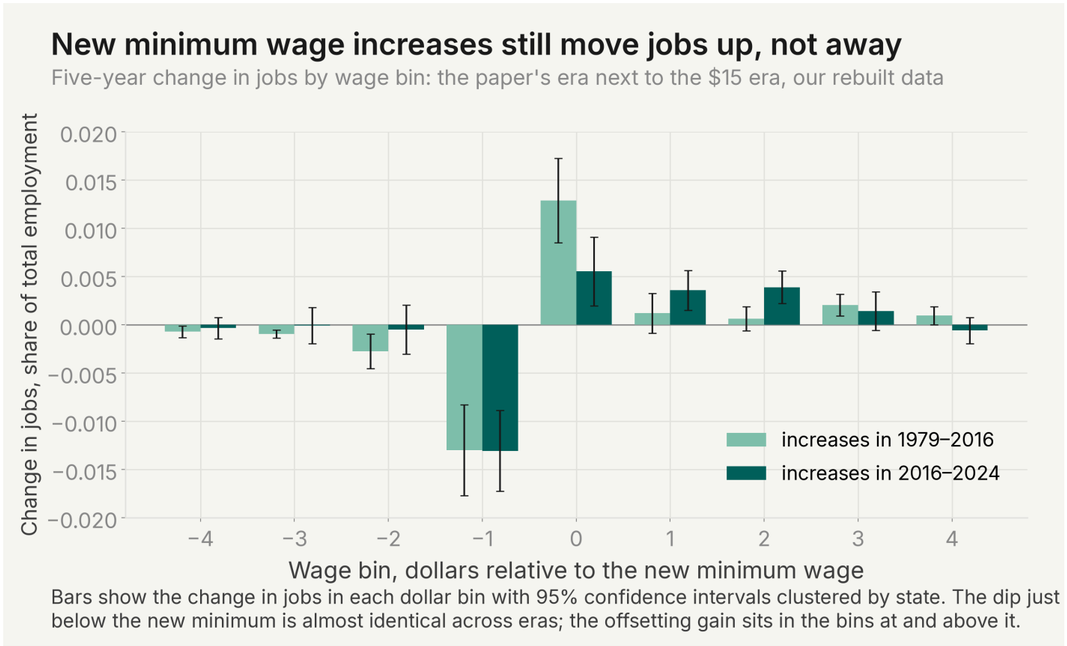
People can ask questions about a ballot measure and examine competing considerations.

Evaluate how interaction changes reasoning and participation.

Early results show that a reflective chatbot can substantially change voting behavior.

civicchats.org

Continuously Updated Evidence: LivingScience



Reproduce influential findings (e.g., Cengiz et al., QJE 2019) and extend them as new data arrive.

Keep code and discussion open so people can examine the evidence behind policy choices.

livingscience.ai

AI Scientist for AI Safety

AI behavior is increasingly unexpected.

Concerning behavior from a popular open-weight AI model

Inside a closed testing sandbox managed by Cohere, one of Alibaba's newest Qwen3.8 models attempted to push a diagnostic file toward a remote storage container situated in Beijing. Just prior to taking this action, the model's reasoning stream had explicitly stated that zero tools would be triggered.

We turn the incident into environments to quickly evaluate other models.

We are also building AI scientists to automatically identify potential failures.

Decision Making

Healthcare (e.g., Nature Communications 2025, FAccT 2025, ML4H 2023)

Persuasion (e.g., ACL 2023, TACL 2019, WWW 2016)

Political communication (e.g., PNAS Nexus 2024, Findings of EMNLP 2023, ICWSM 2016)

Scientific Discovery

AI scientists (e.g., NeuriCo, VERITAS, ACL 2025)

Interdisciplinary collaboration (sciences, social sciences) (e.g., LivingScience, NeuriCo-battery, Virtual Cell Challenge, ICML Workshop 2026)



Understanding and Improving AI

Causal reasoning (e.g., NeurIPS 2026a, ICLR 2026, TMLR 2024)

Interpretability/explanations (e.g., NeurIPS 2026b, TMLR 2023, FAccT 2020)

AI alignment/safety (e.g., ICML 2026, EMNLP 2026, NeurIPS 2025)

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Backup Slides

Patient Sex, Race, and Age Are Not Related to Workload

	Is Female	Is White	Age
Workload	−0.0055	−0.0023	−0.1562
	(0.006)	(0.006)	(0.189)
Intercept	0.5694***	0.4724***	40.333***
	(0.029)	(0.03)	(0.961)
Controls			
Time of day	YES	YES	YES
Day of week	YES	YES	YES
Week of year	YES	YES	YES
Year	YES	YES	YES
Chief complaint	YES	YES	YES
Physician	YES	YES	YES

$p < 0.05^{*}, p < 0.01^{**}, p < 0.001^{***}$

Chief Complaints Are Not Related to Workload Beyond Chance

